

Course Code: CS3005	Course Name: Theory of Automata
Instructor Name: bakhtawar	
Student Roll No:	Section No:

Total Time: 30 minutes

Maximum Points: 35

Question # 1 (5+5 +5 Points)

Write down the RE for the following languages.

- a) Language of the words that start with ab and end aa and have baa as a substring.
- b) Define language such that it would contain at least one double letter
- c) Regular expression for a valid email ID of your student ID.

Question # 4 (5+5+5 Points)

Design the FA for the following languages

- a. $L = \{bbb, aaa, \lambda, b, bb\}$
- b. $L = \{zwv : |w| = 1, v \in \{a, b\}^+, z \in \{a, b\}^*, w \in \{a, b\}^+\}$
- c. Design a DFA that accepts set of strings such that every string contains '00' as a substring but not '000' as a substring.

Question # 5 (5 Points)

Convert the following NFA to DFA.

